Lesson 5.4 Guided Notes

Session 1 · Area of Regular Polygons

I can find the area of a regular polygon by finding the area of one triangle and multiplying. · 6.G.A.1

Word / Spanish	Meaning — complete the blank	Example
regular polygon / polígono regular	All sides _____ and all angles equal.	stop sign, hexagon tile
congruent / congruente	Exactly the same _____ and shape.	all center triangles match
apothem / apotema	Distance from the _____ to the middle of a side.	height of a center triangle

Key idea

Lines from the center to each vertex make _____ triangles.

A regular polygon with n sides makes _____ center triangles.

Area = n × (½ × side × _____).

Worked example

A regular pentagon has sides of 10 cm and an apothem of about 7 cm. Find its area.

5 sides make _____ center triangles.

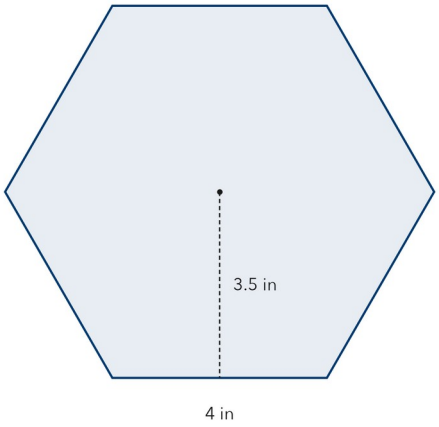
One triangle: ½ × 10 × 7 = _____ cm².

All triangles: 5 × 35 = _____ cm².

Answer: Area ≈ _____ cm²

You try

1. Find the area of the regular hexagon.



_____

2. How many center triangles does each regular polygon make?
a. octagon _____
b. decagon _____

_____

Sum it up: To find the area of a regular polygon, I ___ and then ___.

_____

Session 2 · Composite Figures

I can find the area of a composite figure by adding or subtracting the areas of simpler shapes. · 6.G.A.1

Word / Spanish	Meaning — complete the blank	Example
composite figure / figura compuesta	A shape made of two or more _____ shapes.	rectangle + triangle
decompose / descomponer	To _____ a shape into smaller shapes you know.	L shape: 2 rectangles
subtract / restar	To _____ the area of a piece that is cut out.	wall − window

Key idea

Split a composite figure into rectangles, triangles, or other shapes you _____.

_____ the areas of pieces that are joined together.

_____ the area of a piece that is cut out.

Worked example

A rectangle is 14 cm by 9 cm. A triangle notch with base 6 cm and height 4 cm is cut out. Find the area left.

Whole rectangle: 14 × 9 = _____ cm².

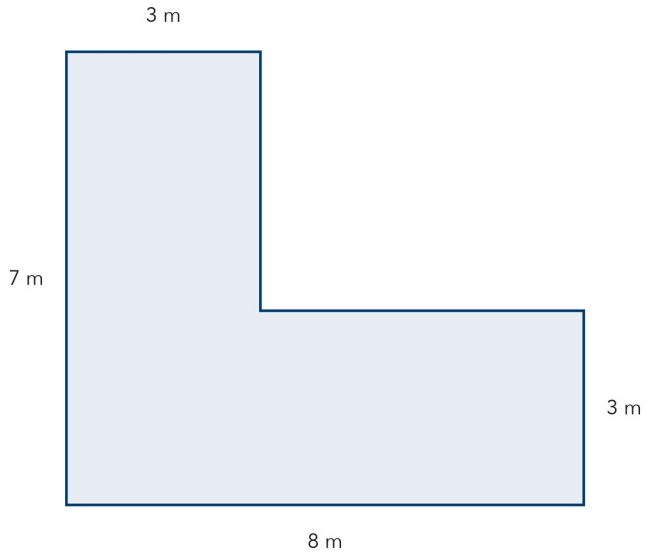
Cut-out triangle: ½ × 6 × 4 = _____ cm².

The triangle is removed, so _____.

Answer: 126 − 12 = _____ cm²

You try

1. Find the area of the L-shaped figure. All corners are right angles.



_____

2. Write add or subtract.
a. a yard, not counting its pool: _____
b. a wall plus its triangle roof: _____

_____

Sum it up: I add areas when ___, and I subtract areas when ___.

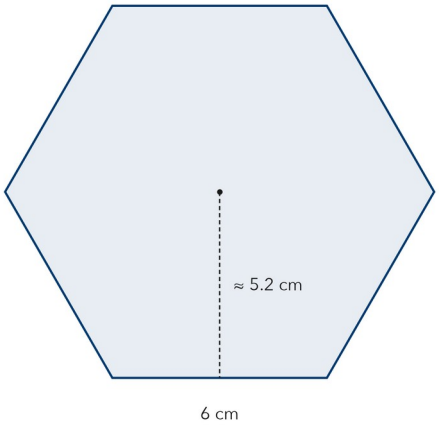
_____

Session 1 · Model, reason, and explain

Area of Regular Polygons · 6.G.A.1

Success: represent the quantities, justify the relationship, and check the result with the correct units.

A regular hexagon has side 6 cm and a supplied apothem approximately 5.2 cm. Draw its six center triangles and estimate area.



_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: uses center-to-vertex length as the apothem or drops approximation notation.. Explain what changes and what stays true.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Composite Figures · 6.G.A.1

Success: represent the quantities, justify the relationship, and check the result with the correct units.

A 12-by-8 cm rectangle has a 4-by-3 cm rectangle removed from a corner. Sketch the remaining shape and find area by subtraction and by two rectangles.



_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: adds a removed region or double-counts overlapping pieces.. Explain what changes and what stays true.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?